

OUTPUTS

Question 1 — Inventory Management System:

Part A — Product class with Array of Objects

```
===== INVENTORY REPORT =====
ID      Name      Price      Qty      Total Value
Product Id: 1001Name: Widget-A Price: 250    Quantity: 120    Total Value: 30000
Product Id: 1002Name: Bolt-B   Price: 15.5     Quantity: 8      Total Value: 124
Product Id: 1003Name: Mouse-C  Price: 500      Quantity: 15     Total Value: 7500
Product Id: 1004Name: Keyboard-DPrice: 800     Quantity: 5      Total Value: 4000
Product Id: 1005Name: Monitor-E Price: 12000    Quantity: 3      Total Value: 36000

Highest Value Product:
Product Id: 1005Name: Monitor-E Price: 12000    Quantity: 3      Total Value: 36000
Enter Threshold: 10
Low Stock Products:
ID      Name      Price      Qty      Total Value
Product Id: 1002Name: Bolt-B   Price: 15.5     Quantity: 8      Total Value: 124
Product Id: 1004Name: Keyboard-DPrice: 800     Quantity: 5      Total Value: 4000
Product Id: 1005Name: Monitor-E Price: 12000    Quantity: 3      Total Value: 36000
```

Part B — Function Overloading

```
Console × Progress Console Eclipse IDE for C/C++
<terminated> (exit value: 0) problem1.B Debug (1) [C/C++ Application]
500
525
590
900
800
```

Question 2 — Hospital Patient Registry

```
[Constructor] Full admission: Meera Joshi
[Constructor] Emergency: Mayur Patil
[Constructor] Default patient registrated

Patient record:
ID : 1001
Name : Meera Joshi
Age : 34
Ward : Cardiology
Blood group : B+
[Constructor] Default patient registrated
[Constructor] Default patient registrated
[Constructor] Default patient registrated
[Constructor] Default patient registrated

Patient record:
ID : 0
Name : Unknown
Age : 0
Ward : General
Blood group : O+

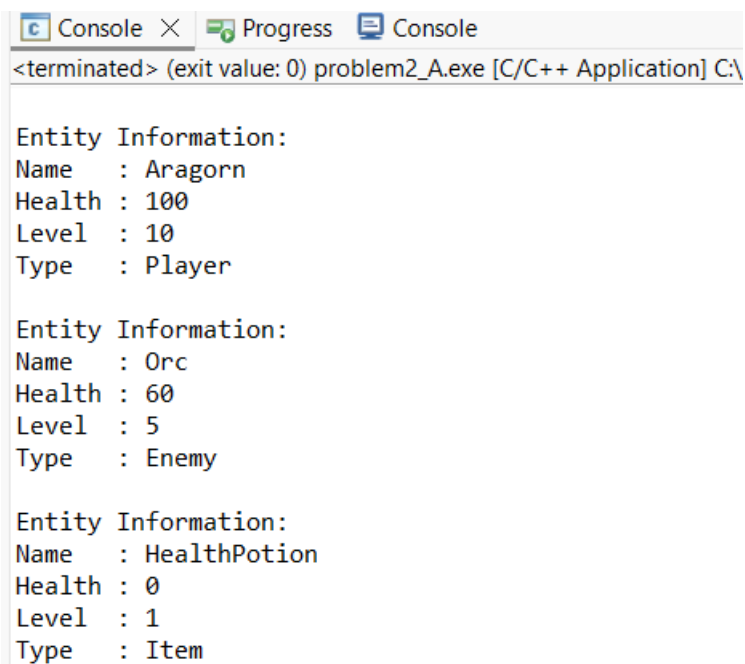
Patient record:
ID : 0
Name : Unknown
Age : 0
Ward : General
Blood group : O+

Patient record:
ID : 0
Name : Unknown
Age : 0
Ward : General
Blood group : O+
```

```
Patient record:
ID : 0
Name : Unknown
Age : 0
Ward : General
Blood group : O+
Ward transfer: Mayur Patil --> ICU
[Destructor]Patient UnknownDischarged
[Destructor]Patient UnknownDischarged
[Destructor]Patient UnknownDischarged
[Destructor]Patient UnknownDischarged
[Destructor]Patient UnknownDischarged
[Destructor]Patient Mayur PatilDischarged
[Destructor]Patient Meera JoshiDischarged
```

Question 3 — Game Engine Entity System

Part A — this Pointer and Method Chaining



The screenshot shows a console window titled "Console" with a "Progress" bar. The output text is as follows:

```
<terminated> (exit value: 0) problem2_A.exe [C/C++ Application] C:\
Entity Information:
Name : Aragorn
Health : 100
Level : 10
Type : Player

Entity Information:
Name : Orc
Health : 60
Level : 5
Type : Enemy

Entity Information:
Name : HealthPotion
Health : 0
Level : 1
Type : Item
```

Part B — Namespaces

```
<terminated> (exit value: 0) problem3_b.exe [  
Physics Clamp: 100  
Physics Lerp: 15  
GameMath Clamp: 100  
GameMath Lerp: 15  
Block Scope Physics Clamp: 100  
Block Scope Physics Lerp: 25  
GameMath Clamp Again: 50
```

Part C — Dynamic 2D Game Map

```
<terminated> (exit value: 0) problem 3_c.exe [C/C++ Application] C:\C++\proble  
Enter number of rows: 3  
Enter number of columns: 4  
  
===== GAME MAP (3 x 4) =====  
4 4 4 0  
2 3 0 0  
1 1 4 2  
  
Legend:  
0 = Grass  
1 = Water  
2 = Mountain  
3 = Forest  
4 = Dungeon  
  
Tile Count:  
Grass : 3  
Water : 2  
Mountain : 2  
Forest : 1  
Dungeon : 4  
  
Local Level : 10  
Global Level : 1  
Playing: sword_clash
```